JUNIOR SCHOOL LEARNING HUB

GRADE 9 MATHEMATICS

CONTINUOUS ASSESSMENT TEST 1 (CAT 1)

TERM 1

Time: 1 hour 15 minutes **Total: 50 marks**

Learner's Name: ..... Grade: 9 Date: .....

INSTRUCTIONS

1. Answer ALL questions.
2. Show all necessary working. Unsupported answers may not earn full marks.
3. Use a calculator only where permitted by your teacher.
4. Write your answers clearly and include correct units where required.

SECTION A: SHORT-ANSWER QUESTIONS (20 MARKS)

1. Evaluate: $-18 + 7 - (-9)$. (2 marks)

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2. Find the cube of -4. (2 marks)

.....

3. Evaluate $\sqrt[3]{(-216)}$. (2 marks)

.....

4. Simplify: $2^3 \times 2^4 \div 2^2$. Give your answer as a power of 2 and as an integer. (2 marks)

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5. Express 0.00056 in standard form. (2 marks)

.....

6. A machine produces 180 items in 6 hours. At the same rate, how many items will it produce in 10 hours? (2 marks)

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7. State the order of the matrix [[3, 5, 1], [2, 0, 4]]. (2 marks)

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8. Given A = [[2, 1], [4, 3]] and B = [[1, 5], [2, 0]], find A + B. (2 marks)

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9. Find the value of $5^0 + 2^3$. (2 marks)

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10. Write 81 as a power of 3. (2 marks)

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SECTION B: STRUCTURED QUESTIONS (30 MARKS)

11. Integers (6 marks)

(a) Arrange the following integers in ascending order: $-3, 7, 0, -9, 4$. (2 marks)

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(b) Evaluate: $-6 \times [4 - (-3)] + 20 \div (-5)$. (2 marks)

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(c) The temperature at 6:00 a.m. was $-4°C$. It rose by $11°C$ and later fell by $6°C$. Find the final temperature. (2 marks)

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12. Cubes, cube roots and indices (6 marks)

(a) Evaluate $7^3 - \sqrt[3]{512}$. (2 marks)

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(b) Simplify, expressing your answer with positive indices: $(3^5 \times 3^2) \div 3^4$. (2 marks)

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(c) Evaluate 2^{-3} and express your answer as a fraction. (2 marks)

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13. Compound proportion and rates of work (6 marks)

(a) 8 workers can complete a task in 15 days. How many days would 12 workers take if they work at the same rate and all workers are equally productive? (3 marks)

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(b) A tap fills 240 litres of water in 8 minutes. Calculate the rate of flow in litres per minute and the amount filled in 15 minutes at the same rate. (3 marks)

.....

14. Matrices (6 marks)

Given $P = [\,[4, -2], [3, 5]\,]$ and $Q = [\,[1, 6], [-2, 4]\,]$:

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(a) State the order of P. (1 mark)

.....

(b) Find $P - Q$. (2 marks)

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(c) Find $2P$. (2 marks)

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(d) State one real-life situation in which information may be organised in a matrix. (1 mark)

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15. Application of indices and standard form (6 marks)

(a) Simplify: $(2^3)^2 \div 2^4$. (2 marks)

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(b) A distance is 4.8×10^5 metres. Write the distance in ordinary form. (2 marks)

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(c) A learner claims that $10^3 + 10^2 = 10^5$. Explain why the claim is incorrect and give the correct answer. (2 marks)

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END OF QUESTION PAPER

JUNIOR SCHOOL LEARNING HUB

GRADE 9 MATHEMATICS – CAT 1

MARKING SCHEME | TERM 1

Total: 50 marks. Award method marks where indicated. Accept mathematically equivalent correct working and answers.

SECTION A: SHORT-ANSWER QUESTIONS (20 MARKS)

1. $-18 + 7 - (-9) = -18 + 7 + 9 = -2$. (2 marks)

2. $(-4)^3 = (-4)(-4)(-4) = -64$. (2 marks)

3. $\sqrt[3]{(-216)} = -6$ because $(-6)^3 = -216$. (2 marks)

4. $2^3 \times 2^4 \div 2^2 = 2^{(3+4-2)} = 2^5 = 32$. (1 mark for power; 1 for integer)

5. $0.00056 = 5.6 \times 10^{-4}$. (2 marks)

6. Rate = 180 ÷ 6 = 30 items/hour; 30 × 10 = 300 items. (2 marks)

7. 2 rows and 3 columns; order 2 × 3. (2 marks)

8. A + B = [[3, 6], [6, 3]]. (2 marks)

9. $5^0 + 2^3 = 1 + 8 = 9$. (2 marks)

10. $81 = 3^4$. (2 marks)

SECTION B: STRUCTURED QUESTIONS (30 MARKS)

11. Integers (6 marks)

- (a) −9, −3, 0, 4, 7. Correct arrangement: 2 marks.
- (b) $-6 \times [4 - (-3)] + 20 \div (-5) = -6 \times 7 - 4 = -42 - 4 = -46$. Correct bracket evaluation: 1 mark; final answer: 1 mark.
- (c) $-4 + 11 - 6 = 1°C$. Correct operation: 1 mark; final answer with unit: 1 mark.

12. Cubes, cube roots and indices (6 marks)

- (a) $7^3 - \sqrt[3]{512} = 343 - 8 = 335$. Correct cube: 1 mark; cube root and answer: 1 mark.
- (b) $3^{(5+2-4)} = 3^3 = 27$. Index law: 1 mark; answer: 1 mark.
- (c) $2^{-3} = 1/2^3 = 1/8$. Reciprocal step: 1 mark; answer: 1 mark.

13. Compound proportion and rates of work (6 marks)

- (a) Workers and days are inversely proportional. 8 × 15 = 12 × d; d = 120/12 = 10 days. Correct inverse relationship: 1 mark; equation/substitution: 1 mark; answer: 1 mark.
- (b) Rate = 240 ÷ 8 = 30 L/min (1 mark). In 15 minutes: 30 × 15 = 450 L (2 marks: method 1, answer with unit 1).

14. Matrices (6 marks)

- (a) 2 × 2. (1 mark)
- (b) P − Q = [[4−1, −2−6], [3−(−2), 5−4]] = [[3, −8], [5, 1]]. Correct corresponding subtraction: 1 mark; matrix: 1 mark.
- (c) 2P = [[8, −4], [6, 10]]. Correct multiplication: 1 mark; matrix: 1 mark.
- (d) Example: arranging learners' attendance by class and day, shop sales by product and week, or sports scores by team and match. (1 mark)

15. Application of indices and standard form (6 marks)

- (a) $(2^3)^2 \div 2^4 = 2^6 \div 2^4 = 2^2 = 4$. Power-of-a-power law: 1 mark; answer: 1 mark.
- (b) 4.8×10^5 = 480,000 metres. (2 marks)
- (c) The law of indices for addition does not allow adding exponents. $10^3 + 10^2 = 1{,}000 + 100 = 1{,}100$. Explanation: 1 mark; correct answer: 1 mark.

Assessment guidance

- Total marks: 50.
- Check that learners show working in structured questions.
- Accept alternative correct methods, including clear unitary-method solutions.
- Do not award a final-answer mark where the answer is incorrect; award valid method marks as indicated.
- Teacher should confirm that all tested sub-strands have been taught before administering the CAT. If the school's Term 1 sequence differs, adjust the scope accordingly.

Prepared by Junior School Learning Hub. Original teacher-prepared assessment; not an official KICD examination.